

10 Reactants P and Q are involved in a chemical reaction.

In an experiment involving a large excess of Q, the graph of [P] against time has a constant half-life.

In further experiments, the rate of reaction is measured using different initial concentrations of Q. The results are shown.

[P] / mol dm ⁻³	[Q] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
0.0200	0.0250	4.76×10^{-4}
0.0200	0.0600	2.74×10^{-3}

What is the numerical value of the rate constant?

A 0.762

B 0.952

C 38.1

D 47.6